

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Agenda 6/30

More on Statements and Structured Programming

- Iteration statements
- Writing simple arithmetic programs
- The elements of structured programming
- References and further readings
- Q&A

120 min (incl. Q & A)



Iteration statements

- Iteration statement

- while statement



```
while (condition)
    statement
```

- for statement



```
for (for-init statement conditionopt; expressionopt) statement
```

- do-while statement

```
do statement while (expression) ;
```

- Each of these statements executes a statement (called the *controlled statement* or the *body of the loop*) repeatedly until the condition becomes false or the programmer breaks out of the loop some other way.

while statement, increment and decrement operators

- The world's first "real program" running on a stored program computer. 
- Write a program to calculate the squares of int values up to 100.

Control information

```
// calculate and print a table of squares 0-99
#include <iostream>
int main()
{
    int i = 0;
    while (i < 100) {
        cout << i << '\t' << i * i << std::endl;
        ++i; // increment i (that is, i becomes i+1)
    }
}
// no it wasn't actually written in C++ ☺
```

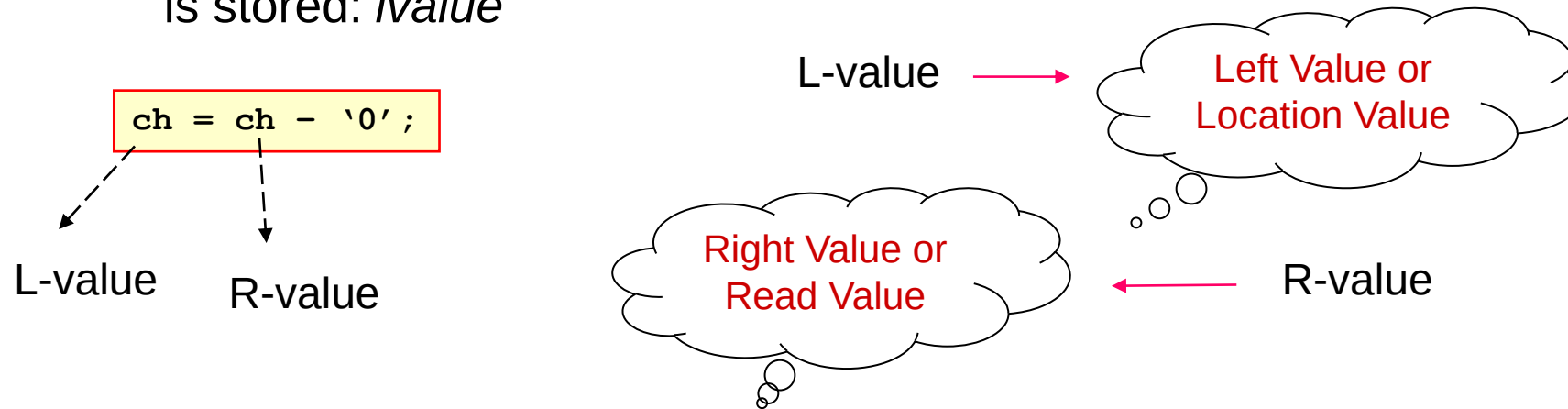
- Increment and decrement operators.

Operator	Function	Use
++	Post increment	lvalue++
--	Post decrement	lvalue--
++	Pre increment	++lvalue
--	Pre decrement	--lvalue

- The operators ++ and -- can be used as both *prefix* and *postfix* operators.

Variable, Object, L-Value and R-Value

- There are two values associated with a symbolic variable:
 - The data value, stored at some location in memory: *rvalue*
 - Its location value; that is, the address in memory at which its data value is stored: *lvalue*



- lvalue is something that can be on the left hand side of an assignment.
- a *variable* is an object that has a name.
- an *object* is a contiguous region of storage.
- It is possible to have objects that do not have name like temporary objects.
- lvalue is an expression that refers to an object.

`*p[a+10] = 7;`

for statement

```
for (for-init statement conditionopt; expressionopt)  
    statement
```

→ for header
→ for body

- Re-write the table of squares of with for statement.

```
#include <iostream>

int main()
{
    for (int i = 0; i < 100; i++) {
        std::cout << i << '\t' << i * i << std::endl;
    }

    return 0;
}
```

```
for-init statement  
while (condition) {  
    statement  
    expression  
}
```

- Write a program that read an integer n from input and computes the summation 1, 2, 3, ... n. then compute $n * (n + 1) / 2$. compare two results.

Program

```
#include <iostream>

int main()
{
    std::cout << "Enter an integer number (0 for exit): ";
    int n;
    cin >> n;

    // to be continued on next page
}
```

for statement_{cont}

```
// continued from last page
if (n == 0) // if (!n)
    return 0;
int sum1 = 0; // must be initialized
for (int i = 1; i <= n; i++) {
    sum1 += i;
}

int sum2 = n * (n + 1) / 2;
std::cout << "sum with first method = " << sum1 << std::endl;
std::cout << "sum with second method = " << sum2 << std::endl;

return 0;
}
```

- In for statement, the *loop variable*, the *termination condition*, and the *expression* that updates the loop variable can be presented “*up front*” on a single line. This can greatly increase readability and thereby decrease the frequency of errors.
- Declarations in for statements: A variable can be declared in the initializer part of a for statement.

Declarations in for statements

```
{  
  ↑ int i; // decl., def., uninitialized  
  for (i = 1; i <= n; ++i) { // assignment  
    ↑ // ...  
  }  
}
```

just Assignment

- Scope of i: Outside of the for statement. i is not local.

decl., def. & init.

```
for (int i = 1; i <= n; ++i) { // good  
  ↑  
}
```

- Scope of i: until the end of for statement. i is **local**.
- If the final value of an index needs to be known after exit from a for loop, the index variable must be declared outside the for loop.



Language-technical rule:
Locality is good.

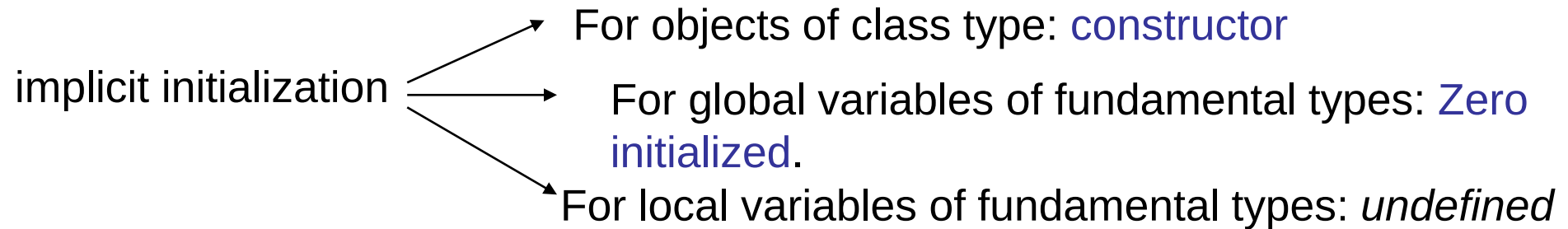
Initialization, default initialization

- **Initialization**: giving an object an initial value.

```
int sum1 = 0;
```

- When we do not specify an initial value for a variable we are implicitly relaying on **default initialization**.

default initialization → implicit initialization

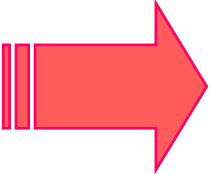


```
int i; // i = 0;
double d; // d = 0.0
int main()
{
    std::string s; // s is initialized by constructor
    int k; // k does not have well-defined value
    bool b = false; // explicitly initialized
    char c = 'a'; // explicitly initialized
}
```

Simple computation, simple programs

- Compute $n!$: Get n from input and compute $n * (n - 1) * (n - 2) * \dots * 1$. Note that: $0! = 1$ and $1! = 1$. Handle input for negative numbers.

Program



```
#include <iostream>
int main()
{
    std::cout << "Enter an integer number: (-1 for exit) ";
    int n;
    std::cin >> n;
    if (n == -1) {
        return 0;
    }
    if (n == 0) {
        std::cout << n << "! = " << 1 << '\n';
        return 0;
    }
    long int Fact = 1; // must be explicitly initialized
    for (int i = n; i > 0; --i) {
        Fact *= i;
    }
    std::cout << n << "! = " << Fact << '\n';

    return 0;
}
```

R references and further readings

- [Str00] Bjarne Stroustrup. The C++ Programming Language (Special Edition). Addison Wesley. Reading Mass. USA. February 2000.
- [Str08] Bjarne Stroustrup. Programming: Principles and Practice in C++, 1st edition, Addison Wesley, December 2008.
- [Str13] Bjarne Stroustrup. The C++ Programming Language. 4th edition. Addison Wesley. Reading Mass. USA. May 2013.
- [Str22] Bjarne Stroustrup. A Tour of C++, 3rd edition, Addison Wesley, September 2022.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

